NuMicro® Family

Arm® Cortex®-M0-based Microcontroller

M031 Series CMSIS BSP
Revision History

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Revision 3.10.000 (Released 2026-10-15)

1. Fix I2S PDMA NAU8822 sample code build error
2. Fix Doxygen warnings, comments, typos, and AStyle issues
3. Modify some drivers to fix the MISRA violation.
4. Modify some drivers to fix the MISRA violation.
5. Updated startup code to support weak ProcessHardFault
6. Merge changes Id0cc2b3d,I28d87ffb
7. Merge "Fix ISP sample build fail."
8. Merge "Modify ebi.c to fix the MISRA violation"
9. Modify gpio.c to fix the GPIO related MISRA violation
10. Modify timer.h to fix the TIMER related MISRA violation
11. Modify timer.h/timer.c/clk.h to fix the Timer related MISRA violation
12. Fix ISP sample build fail.
13. Modify ebi.c to fix the MISRA violation
14. Merge "[SBOM] Strengthen artifact and license evidence validation"
15. [SBOM] Strengthen artifact and license evidence validation
16. Merge "Modify hdiv.h and spi.h to fix MISRA violation"
17. Modify hdiv.h and spi.h to fix MISRA violation
18. [SBOM][M031] Sync formal release V3.07.000-71-g4dc3d5af
19. [SBOM][M031] Refresh current SBOM evidence
20. Merge "[SBOM][M031] Sync formal release V3.07.000-60-g03e4dea8"
21. [SBOM][M031] Sync formal release V3.07.000-60-g03e4dea8
22. [SBOM][M031] Complete formal release metadata
23. Merge "[SBOM][M031] Complete formal release metadata"
24. Merge "M031: Move GCC HardFault handler from retarget to startup for consistent toolchain handling"
25. Merge "Fix MISRA-C violations for I2C, PDMA, UART, QSPI and USCI driver."
26. Fix MISRA-C violations for I2C, PDMA, UART, QSPI and USCI driver.
27. M031: Move GCC HardFault handler from retarget to startup for consistent toolchain handling
28. Update Product and TestSample SBOMs
29. Merge "Update Product and TestSample SBOMs"
30. Modify bpwm/pwm/ebi/spi/hdiv drivers to fix some MISRA violation.
31. Merge "Modify bpwm/pwm/ebi/spi/hdiv drivers to fix some MISRA violation."
32. Update root SBOM artifacts to V3.07.000-57-g822260b
33. Merge "[SBOM][M031] Add TestSample component hash metadata"
34. [SBOM][M031] Add TestSample component hash metadata
35. Fix CMSIS hash metadata and preserve SBOM files
36. Merge "Fix CMSIS hash metadata and preserve SBOM files"
37. Update startup, system, and retarget files
38. Correct RTC reference definition for frequency compensation
39. Fix RTC/USBD/WDT MISRA C violations
40. Update FMC driver for MISRA-C compliance
41. Update SBOM files
42. Update SBOM files
43. Add initial M031 BSP SBOM and SCA evidence
44. Normalize HIDCtrlTransferTest solution line endings
45. Move NuBLE Android app to Tool directory
46. Add readme file for BLE
47. Preserve Nu_DFU INF file byte-for-byte

48. Move BLE host tools to Tool directory & Add USBD Windows Tool Readme
49. Normalize HIDTransferTest solution line endings
50. Modify for SBOM
51. Remove semihost VSCode project
52. Update SBOM files
53. Add SBOM files
54. Fix FMC multi-word programming 1. Remove redundant code 2. Correct return value handling on timeout
55. Fix FMC Multi-Word Program 1. Fix BSP driver missing status check for the final 4-word write. 2. Correct the retransmission address calculation for timeout recovery.
56. Modify PDMA EnableInt function to support two more mask at the same time
57. Fix building issues on Linux.
58. Fix GITHUB ACTION GCC/VSCode Build Error
59. Fix GITHUB ACTION VSCode Build Error
60. Add VSCode Quick Start Guide.pdf
61. Update VSCode project files
62. Modify to support cortex-debug-nuvton and GCC 14.3.1

Revision 3.09.000 (Released 2025-08-12)

1. Fixed the Keil project so that NU_LINK debugger is selected by default.
2. Revised i2c/ui2c driver and fixed ISP_I2C/ISP_SPI sample.

Revision 3.08.000 (Released 2025-06-07)

1. Fix typo.
2. Update spi driver.
3. Add Add annotations to remind users to unlock system register before using FMC.
4. Modify USB HID sample code to send NAK instead of 0 to fix host can't sleep issue.
5. Add VSCode project to each sample.
6. Update PDMA driver - the order of settings int Scatter-gather mode.
7. Upgrade CMSIS to 6.1.0 2. update project include path for CMSIS 6.1.0.
8. Add timeout to I2C slave sample code.
9. Update CLK_ClockDetector sample code for PF2 (XT1_OUT) and PF3 (XT1_IN).

Revision 3.07.000 (Released 2024-10-16)

1. Added I^2C error condition process.
2. Added .uvprojx project file for Keil5.
3. Removed .uvproj file for Keil4.
4. Minor changes for sample code.
5. Minor bug fix.

Revision 3.06.000 (Released 2023-11-14)

1. Added PWM_Brake and ADC_SingleCycleScanMode2 samples.
2. Added LED status for USBD samples.
3. Fixed USB MSC sample access fail issue.
4. Fixed Keil Compiler 6 issue.
5. Minor changes for sample code.
6. Minor bug fix.

Revision 3.05.000 (Released 2022-03-18)

1. BLE supported GCC and IAR toolchain.
2. Removed emWin samples and library.
3. Minor changes for sample code.
4. Minor bug fix.

Revision 3.04.000 (Released 2021-12-28)

1. Supported BLE for M031BT/M032BT.
2. Added I2C_SMBus_Master and I2C_SMBus_Slave samples.
3. Added Sample codes for FreeRTOS V2020.12.
4. Added ISP_MSC sample.
5. Added Cache invalidation for FMC Write related functions.
6. Added new API for TIMER.
7. Updated I^2C driver to check STOP bit when I^2C transmission completed.
8. Updated HID samples polling interval to 10.
9. Updated samples to avoid infinite loop.
10. Fixed the missing UART6/UART7 in UART_Open function.
11. Minor changes for sample code.
12. Minor bug fix.

Revision 3.03.000 (Released 2020-09-14)

1. Added Apache-2.0 license declaration into BSP.
2. Added fmc_ap_main.uvproj Keil project in FMC_IAP sample.
3. Updated preferences.ini for Eclipse V1.01.016 or later.
4. FMC_IAP sample uses hardware reset instead of software reset.
5. Fixed HIRC trim setting in USBD_Audio_NAU8822_Headset, USBD_Audio_NAU8822_Microphone, and USBD_Audio_NAU8822_Speaker samples.
6. Fixed I^2C wakeup done flag checking fail issue.
7. Minor changes for sample code.
8. Minor bug fix.

Revision 3.02.000 (Released 2019-10-4)

1. Supported GPIOE, GPIOG, GPIOH, UART3/4/5/6/7, USCI-1, BPWM, QSPI and RTC for new M031 part numbers.
2. Removed I2C_Master_PDMA, I2C_PDMA, I2C_Slave_PDMA and USBD_Audio_NAU8822 samples.
3. Added ISP related samples.
4. Added BPWM, QSPI and RTC related samples.
5. Added emWin sample codes and library.
6. Updated I2C_PDMA_TRX, USCI_SPI_PDMA_LoopTest, USCI_I2C_Loopback_10bit, USCI_I2C_Loopback, FMC_CRC32, FMC_ExeInSRAM, FMC_MultiBoot, FMC_MultiWordProgram, FMC_ReadAllOne, USBD_Audio_NAU8822_HeadSet, USBD_Audio_NAU8822_Microphone and USBD_Audio_NAU8822_Speaker samples.
7. Minor changes for sample code.
8. Minor bug fix.

Revision 3.01.000 (Released 2019-04-12)

1. Added USBD_Audio_NAU8822 sample for UAC class.
2. Removed I2C_Double_Buffer_Slave sample.

3. Updated default stack size from 1KB to 512B.
4. Updated definition from SYS_xx_ADC_xx to SYS_xx_ADC0_xx.
5. Updated samples for M031EB0AE, M031FB0AE and M031TB0AE.
6. Minor changes for sample code.
7. Minor bug fix.

Revision 3.00.000 (Released 2018-06-29)

1. Preliminary release version.

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